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1. In equilibrium condition of mutarotation, the percentages of α and β anomers which of the following respectively?
 - a. $\alpha \frac{1}{2}$, $\beta \frac{1}{2}$
 - b. $\alpha \frac{2}{3}$, $\beta \frac{1}{3}$
 - ☒ c. $\alpha \frac{1}{3}$, $\beta \frac{2}{3}$
 - d. $\alpha \frac{1}{4}$, $\beta \frac{3}{4}$
2. Which one of the following structure is heparin?
 - a. Glucuronic acid- N-acetyl glucosamine
 - ☒ b. Iduronic acid- N-acetyl glucosamine
 - c. Glucuronic acid- N-acetyl galactosamine
 - d. Iduronic acid -N acetyl galactosamine
3. Which one of the following sugars is Aldose?
 - a. Ribulose
 - b. Dihydroxy acetone
 - c. Fructose
 - ☒ d. Galactose
4. Which one of the following sugars has 6 carbon atoms?
 - ☒ a. Arabinose
 - b. Surbose
 - c. Xylose
 - d. Xylolose
5. Which one of the following is N-acetyl Neuraminic Acid?
 - a. Music acid
 - b. Saccharic acid
 - c. Mannaric acid
 - ☒ d. Sialic acid
6. Which one of the following reduced sugars is Sorbitol?
 - a. Ribitol
 - b. Mannitol
 - ☒ c. Glucitol
 - d. Xylitol



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19- How much is the oxygen saturation of blood leaving the tissues (venous blood)?

- a. 0%
- b. 40%
- c. 60%
- d. 100%

20- Hemoglobin carrying CO₂ in blood is called

- a. Oxyhemoglobin
- b. Protonated hemoglobin
- c. Carbaminohemoglobin
- d. Deoxyhemoglobin

21- For hemoglobin, O₂ is:

- a. Homotropic inhibitor
- b. Heterotropic inhibitor
- c. Homotropic activator
- d. Heterotropic activator

22- Which of the following is **C**ephalin?

- a. Phosphatidyl Inositol
- b. Phosphatidyl Choline
- c. Phosphatidyl Serine
- d. Phosphatidyl Ethanolamine

23- Which of the following is synthesized from Arachidonic acid?

- a. Cholesterol
- b. Triglyceride
- c. Lecitin
- d. Leukotriene

24- Which lipoprotein can transfer endogenous lipid in the blood?

- a. Chylomicron
- b. VLDL
- c. LDL
- d. HDL



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30. In the study of enzymes, a sigmoid plot of substrate concentration ($[S]$) versus reaction velocity (V) may indicate:
- Michaelis-Menten kinetics
 - Myoglobin binding to oxygen
 - Competitive inhibition
 - ☒ Cooperative inhibition
31. Which one of the following proteolytic enzymes is activated by acid hydrolysis of the proenzyme form?
- ☒ Pepsin
 - Chymotrypsin
 - Trypsin
 - Elastase
32. All the following correctly describe the active site of an enzyme, EXCEPT:
- It is small relative to the entire enzyme
 - Specificity is defined by the arrangement of certain atoms
 - ☒ It is two-dimensional in structure
 - It is usually a crevice or cleft
33. Which of the following correctly describes allosteric enzymes?
- Regulatory molecules bind the active site
 - ☒ Binding of substrate to one site can affect other sites
 - Regulatory molecules alter equilibrium but not activity
 - Hyperbolic plots are obtained when reaction velocity is plotted against substrate concentration
34. The coenzyme _____ is required to incorporate the methyl group into thymidine, a necessary prerequisite for the production of DNA.
- Biotin
 - ☒ N^5,N^{10} tetrahydrofolate
 - Pyridoxal phosphate
 - Pantothenic acid
35. Vit. C is required for the production and maintenance of
- ☒ Collagen
 - Hormone
 - Ascorbic acid
 - Red blood cells

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36. Spending 15 minutes in sunlight 3 times a week without sunscreen helps increase levels of which vitamin?
- a. Vit C
 - ☒ b. Vit D
 - c. Vit B
 - d. Vit A
37. Vit B12 deficiency caused by lack of intrinsic factor is called:
- ☒ a. Pernicious anemia
 - b. Poor circulation of red blood cells
 - c. Beriberi
 - d. Pellagra
38. Which is correct about NAD⁺?
- a. NAD is a derivative of Vitamin B2
 - b. Can accept two hydrogen atoms
 - ☒ c. Contains a pyridine ring
 - d. Deficiency causes scurvy
39. Which vitamin participates in transamination reactions?
- a. Thiamine
 - b. Riboflavin
 - ☒ c. Pyridoxine
 - d. Cobalamin
40. Biotin attached to lysine is called?
- a. Biotinidase
 - ☒ b. Biocytin
 - c. Apocarcboxylase
 - d. Holocarcboxylase
41. A common feature of thiamine, riboflavin and niacin is that:
- a. They are used in synthesis of blood clotting proteins
 - ☒ b. They all work as coenzyme in energy metabolism
 - c. they help strengthen blood vessels
 - d. They all stabilize cell membranes
42. The deficiency disorder associated with low levels of vitamin B1 is
- a. Dryness of skin
 - ☒ b. Beri beri
 - c. Night blindness
 - d. Neural tube defects

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48. Which is the function of Shine Dalgarno sequence?
- a. Template for protein synthesis
 - ☒ b. Recruiting of ribosome to the initiation site
 - c. Stop codon recognition
 - d. Reduces the efficiency of translation
49. Nonsense mutation is referred to those mutation
- a. that change UAA to UGA
 - ☒ b. that change AAA to UGA
 - c. delete one nucleotide
 - d. insert one nucleotide
50. The antibiotic Rifampin inhibits
- a. Eukaryote transcription
 - ☒ b. Prokaryote transcription
 - c. Eukaryote translation
 - d. Prokaryote translation
51. During the process of protein synthesis, the enzyme responsible for peptide bond formation is?
- ☒ a. peptidyl transferase
 - b. peptidyl translocase
 - c. trans peptidase
 - d. helicase
52. The Sigma factor in prokaryote is responsible for...
- a. termination of RNA chain
 - ☒ b. initiation of RNA chain
 - c. termination of protein chain
 - d. initiation of protein chain
53. Ribozymes
- ☒ a. Are catalytic RNA
 - b. Catalyze ribosome synthesis
 - c. Remove exons from hnRNA
 - d. Responsible enzyme in ribosomes degradation

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54. Posttranscriptional regulation in eukaryotes include all these **EXCEPT**:

- a. splice-site choice
- b. polyA-site choice
- c. mRNA editing
- ☒ d. histon modification

55. All are true about enhancers in eukaryotes except.....

- a. They can be downstream from the promoter
- ☒ b. They are in the coding region
- c. They can be upstream from the promoter
- d. Can be thousands of base pairs from the gene it regulates

56. What are found only in the lagging strand?

- a. Primers
- b. RNA polymerase
- c. DNA polymerase
- ☒ d. Okozaki fragments

57. Which enzyme has proofreading activity?

- a. RNA polymerase
- ☒ b. DNA polymerase
- c. Topoisomerase
- d. Helicase